Reinforcement Learning for Inventory-Aware Market Making in a Synthetic Market

Interim Report

Johnathan Nomman Sheikh Chenhongshu Yu Yating Zhang

1 State of the Project

This project aims to develop a reinforcement learning (RL) agent that can learn an adaptive quoting policy for market making. The objective is to achieve a better tradeoff between profitability and inventory risk than a fixed-spread market maker (the baseline). We have developed a framework consisting of four key components: a synthetic market simulator that mimics the environment (Section 2.1), the RL methodology that learns the policy (Section 2.2), the reward and terminal liquidation mechanism that provides rewards and penalties (Section 2.3), and the evaluation metrics used to assess the agent's performance and alignment with the project's objectives (Section 2.4).

The next section of the report describes the notional architecture. Section 3 outlines the baseline strategy. Section 4 presents preliminary results. Section 5 concludes this report and discusses the next step.

2 Notional Architecture

2.1 Synthetic Market Simulator

Rather than using a fixed historical dataset, we will generate training experience through a synthetic market simulator. The simulator is designed to capture the main mechanisms needed for our research question without reconstructing a full limit order book.

Mid-price dynamics. The mid-price follows a zero-drift arithmetic Brownian motion in discrete time:

$$S_{t+1} = S_t + \sigma\sqrt{\Delta t}\,\epsilon_t, \qquad \epsilon_t \sim \mathcal{N}(0,1). \tag{1}$$

Thus,

$$\mathbb{E}[S_{t+1} - S_t] = 0.$$

The agent therefore receives no predictable directional price signal. Brownian price movements instead create risk when the market maker carries nonzero inventory.

Order flow. Incoming buy and sell market orders are generated independently:

$$N_t^{\text{buy}} \sim \text{Poisson}(\lambda_b \Delta t), \qquad N_t^{\text{sell}} \sim \text{Poisson}(\lambda_s \Delta t). \tag{2}$$

For the core experiments,

$$\lambda_b = \lambda_s,$$

so there is no systematic directional order-flow imbalance.

Quotes and execution. At every time step, the market maker places quotes relative to the current mid-price:

$$P_t^{\text{bid}} = S_t - \tau d_t^{\text{bid}}, \qquad P_t^{\text{ask}} = S_t + \tau d_t^{\text{ask}}, \tag{3}$$

where τ is the tick size and d is the quote distance in ticks.

Conditional on an incoming market order, execution probability decreases exponentially with quote distance:

$$p_{\text{fill}}(d) = e^{-\alpha d}. \tag{4}$$

A buy market order may execute against our ask, while a sell market order may execute against our bid:

$$F_t^{\text{ask}} \mid N_t^{\text{buy}} \sim \text{Binomial}\left(N_t^{\text{buy}}, e^{-\alpha d_t^{\text{ask}}}\right), \tag{5}$$

$$F_t^{\text{bid}} \mid N_t^{\text{sell}} \sim \text{Binomial}\left(N_t^{\text{sell}}, e^{-\alpha d_t^{\text{bid}}}\right). \tag{6}$$

This creates the main market-making tradeoff: quotes closer to the mid-price receive more fills but capture less spread, while wider quotes receive fewer fills but earn more spread when executed.

Inventory and cash are updated by

$$q_{t+1} = q_t + F_t^{\text{bid}} - F_t^{\text{ask}}, \tag{7}$$

$$X_{t+1} = X_t - F_t^{\text{bid}} P_t^{\text{bid}} + F_t^{\text{ask}} P_t^{\text{ask}}. \tag{8}$$

Mark-to-market wealth during the episode is

$$W_t = X_t + q_t S_t. \tag{9}$$

We will impose a common hard inventory limit

$$-q_{\max} \le q_t \le q_{\max}$$

for all strategies. A strategy at $q_{\max}$ cannot execute additional inventory-increasing bid trades, and the corresponding restriction applies at $-q_{\max}$.

The initial simulator settings are:

Parameter	Initial Value	Meaning
S_0	100	Initial mid-price
τ	0.01	Tick size
σ	1 tick/$\sqrt{\text{step}}$	Price volatility
λ_b, λ_s	1 per step	Order-arrival rates
α	0.5	Fill-probability decay
T	200 steps	Episode horizon
$q_{\max}$	20 units	Inventory limit
Quote distances	$\{1, 2, 3\}$ ticks	Discrete actions
c	1 tick/unit	Terminal liquidation cost

These are initial normalized choices and may be adjusted after simulator verification.

2.2 MDP Formulation and RL Strategy

We formulate the market-making problem as a Markov Decision Process (MDP). The core state is

$$s_t = \left[\frac{q_t}{q_{\max}}, \frac{T-t}{T}\right]. \tag{10}$$

The first component measures the current inventory exposure, while the second indicates how much time remains before forced liquidation. Since the mid-price follows a zero-drift Brownian motion with fixed volatility, recent price changes do not contain predictive information about future returns.

The stochastic price process, order arrivals, and executions implicitly define

$$P(s_{t+1} \mid s_t, a_t).$$

Because our learning approach is model-free, this transition probability does not need to be explicitly constructed.

For the core implementation,

$$d_t^{\text{bid}}, d_t^{\text{ask}} \in \{1, 2, 3\}, \tag{11}$$

giving nine discrete quoting actions.

We will use a **Q-learning-based approach**. We may begin with tabular Q-learning using a discretized state representation, or use a neural-network approximation (DQN) when retaining continuous-valued state features.

If time permits, we will extend the quoting decisions to a continuous action space,

$$d_t^{\text{bid}}, d_t^{\text{ask}} \in [d_{\min}, d_{\max}],$$

and explore PPO for continuous-action quoting.

2.3 Reward and Penalty

The agent should earn trading profit while limiting inventory exposure. During the episode, the reward is

$$r_t = (W_{t+1} - W_t) - \zeta \left(\frac{q_{t+1}}{q_{\max}} \right)^2. \tag{12}$$

The first term is mark-to-market PnL. The second is a quadratic inventory penalty, so large long or short positions receive increasingly large penalties.

At the end of the episode, all remaining inventory is forcibly liquidated. Rather than assuming costless liquidation at the mid-price, we apply a fixed liquidation spread. If c is the cost per remaining unit, terminal wealth is

$$W_T^{\text{liq}} = X_T + q_T S_T - c|q_T|. \tag{13}$$

Equivalently, a long position is sold at $S_T - c$, while a short position is covered at $S_T + c$. We initially set c equal to one tick. Normal trading does not include additional transaction fees, slippage, or market impact. The terminal liquidation cost simply gives the agent an economic incentive to reduce inventory as the episode approaches its end.

2.4 Evaluation Metrics

All strategies will be evaluated using the same market parameters and held-out random seeds. The primary validation metrics are:

- **Average episode PnL**, measuring profitability;
- **RMS inventory**,

$$\sqrt{\frac{1}{T} \sum_{t=1}^{T} q_t^2},$$

  measuring inventory exposure and placing greater weight on large inventory excursions;
- **Fill ratio**, measuring the proportion of available order flow executed by the market maker.

RMS inventory is particularly appropriate because, under Brownian price movements, the variance of inventory PnL is proportional to q_t^2.

A successful RL policy need not necessarily produce the highest raw PnL. Comparable PnL with substantially lower inventory exposure or terminal liquidation cost would indicate a better profitability–risk tradeoff.

2.5 Libraries

The simulator will be implemented in **Python** using **NumPy** and wrapped as a **Gymnasium** environment. Tabular Q-learning can be implemented directly, while DQN and the optional PPO extension may use **PyTorch** and/or **Stable-Baselines3**. Pandas and Matplotlib will be used for analysis and visualization.

If the core implementation is completed successfully, an optional robustness experiment will test the learned strategy under a different volatility level or asymmetric order-flow regime.

3 Baseline Strategies

The primary baseline is a **fixed-spread market maker**. At every step it quotes

$$d_t^{\text{bid}} = d_t^{\text{ask}} = 2 \text{ ticks}. \tag{14}$$

Therefore, as the simulated mid-price moves, the baseline continuously updates its prices to

$$P_t^{\text{bid}} = S_t - 2\tau, \qquad P_t^{\text{ask}} = S_t + 2\tau.$$

It does not change its quote distances in response to inventory or remaining time, except for the common hard inventory constraint.

If time permits, we will add an **inventory-aware heuristic baseline**:

$$d_t^{\text{bid}} = d_0 + \beta \frac{q_t}{q_{\max}}, \tag{15}$$

$$d_t^{\text{ask}} = d_0 - \beta \frac{q_t}{q_{\max}}, \tag{16}$$

with values clipped to the allowed range. Positive inventory therefore widens the bid and tightens the ask, while negative inventory produces the opposite behavior.

4 Preliminary Results

First, we verified that the simulator behaves as intended. Specifically, We tested cash and inventory accounting, confirmed that empirical Poisson arrival rates are close to their specified values, verified that fill probability decreases with quote distance, and confirmed that Brownian price increments have approximately zero mean and sufficient variance.

5 Challenges and Next Step

The main challenges that we have encountered include

In the next step, we will train the RL agent and compare its performance with the baseline agent. We will also examine whether the learned policy exhibits interpretable inventory-management behavior, such as becoming less aggressive when buying and more aggressive when selling when inventory is positive.

References

[1] M. Avellaneda and S. Stoikov, "High-Frequency Trading in a Limit Order Book," *Quantitative Finance*, vol. 8, no. 3, pp. 217–224, 2008.

[2] S. Ganesh, N. Vadori, M. Xu, H. Zheng, P. Reddy, and M. Veloso, "Reinforcement Learning for Market Making in a Multi-Agent Dealer Market," arXiv:1911.05892, 2019.

[3] Z. Wang, C. Ventre, and M. Polukarov, "Multi-Agent Reinforcement Learning for Market Making: Competition without Collusion," *Proceedings of the 6th ACM International Conference on AI in Finance*, pp. 814–822, 2025.

[4] T. Spooner, J. Fearnley, R. Savani, and A. Koukorinis, "Market Making via Reinforcement Learning," *Proceedings of the 17th International Conference on Autonomous Agents and Multiagent Systems*, pp. 434–442, 2018.